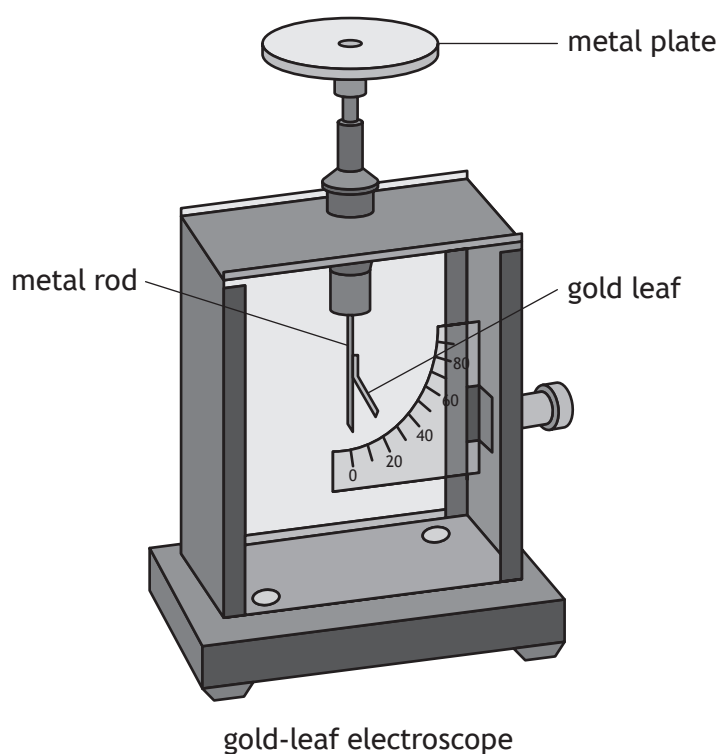


7. A student uses a gold-leaf electroscope to investigate the photoelectric effect. A deflection of the gold leaf on the electroscope shows that the metal plate is charged.

The student charges the metal plate on the electroscope and the gold leaf is deflected.



- (a) Ultraviolet light is shone onto the negatively charged metal plate. The gold-leaf electroscope does not discharge. This indicates that photoelectrons are not ejected from the surface of the metal.

Suggest one reason why photoelectrons are not ejected from the surface of the metal.

1

7. (continued)

- (b) The student adjusts the experiment so that the gold-leaf electroscope now discharges when ultraviolet light is shone onto the plate.

The work function for the metal plate is $6.94 \times 10^{-19} \text{ J}$.

- (i) State what is meant by a *work function of $6.94 \times 10^{-19} \text{ J}$* .

1

- (ii) The irradiance of the ultraviolet light on the metal plate is reduced by increasing the distance between the gold-leaf electroscope and the ultraviolet light source.

State what effect, if any, this has on the maximum kinetic energy of the photoelectrons ejected from the surface of the metal.

2

Justify your answer.

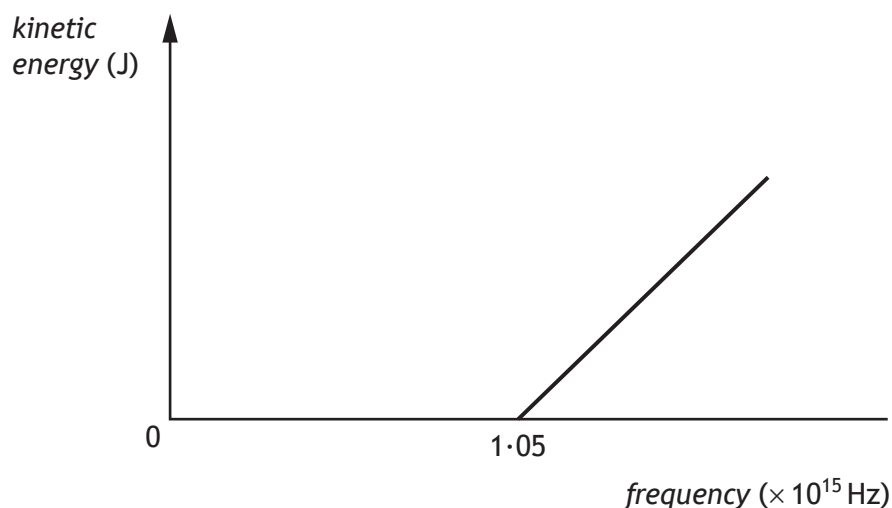
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7. (continued)

- (c) The graph shows how the kinetic energy of the photoelectrons ejected from the metal plate varies as the frequency of the incident radiation increases.

The threshold frequency for the metal plate is 1.05×10^{15} Hz.



The metal plate is now replaced with a different metal plate made of aluminium.

The aluminium has a threshold frequency of 0.99×10^{15} Hz.

Add a line to the graph to show how the kinetic energy of the photoelectrons ejected from the aluminium plate varies as the frequency of the incident radiation increases.

2

(An additional graph, if required, can be found on *page 45*.)

- (d) Explain why the photoelectric effect provides evidence for the particle nature of light.

1



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